



NexusEngine Guide

Core architecture, composition, integration, and migration reference

Version 0.0.4

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Start Here

NexusEngine is the reusable Core beneath games, simulations, editors, and agents. It supplies deterministic runtime primitives and universal semantic capabilities. It does not supply every useful feature.

The shortest correct rule is:

NexusEngine	universal Core behavior
NexusEngine-Kits	reusable optional behavior
Experiments and games	authored product behavior
Hosts and adapters	platform and vendor implementations

Who This Guide Is For

- Application developers who need to assemble a runtime.
- Kit authors who need to choose an owner and dependency contract.
- Host authors who connect browsers, renderers, storage, networks, or SDKs.
- Agents that inspect and plan compositions through MCP.
- Maintainers migrating code to the 0.0.4 semantic Domain model.

Reading Paths

Build something: read First Runtime, Domains And Atomic Kits, then Composition And Recipes.

Integrate a platform: read Registries And Security, then Hosts, Providers, And Adapters.

Use an agent: read Composition And Recipes, then MCP Agent Workflow.

Maintain Core: read State, Lifecycle, And Idempotence, Testing And Proof, and Migrating To 0.0.4.

The Non-Negotiable Boundary

A Core atom has one responsibility, is product-neutral, installs idempotently, owns explicit state, supports snapshot and reset, isolates nondeterminism, and has direct proof. If any requirement is unknown, the behavior remains outside Core until proved.

This fail-closed rule keeps Core small enough to understand and flexible enough to compose.

Sources Of Truth

1. Domain manifest v2 records and the source they reference.
2. Generated catalog, package exports, ownership ledger, and registry hash.
3. This modular guide.
4. Migration and extraction ledgers.

Historical plans, generated run packets, and the retired ProtoKit repository are evidence. They do not override the active manifest catalog.

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The NexusEngine Mental Model

NexusEngine separates meaning from implementation. A Domain names a semantic owner. An atomic Kit installs one capability. A provider performs work behind a contract. An adapter translates between boundaries. A recipe selects and configures atoms without becoming a new runtime owner.

The Vocabulary

Term	Meaning	Example
Engine	Runtime container for state, systems, events, resources, and installed Kits	<code>createEngine()</code>
Domain	Semantic ownership boundary	<code>n:object</code> , <code>n:simulation:motion</code>
Atomic Kit	One installable, idempotent responsibility	Object registry or placement
Provider	Replaceable implementation of a Domain contract	Physics backend or speech synthesis
Adapter	Translation across two public contracts	Motion plan to physics drive request
Recipe	Data selecting Domains, Kits, and settings	Object plus Placement composition
Registry	Non-executable metadata describing exact sources	Package, version, commit, integrity
Host	Application-owned lifecycle and platform boundary	Editor, browser application, server

State Has One Owner

Every durable concept has one state owner. Other Domains refer to that state through descriptors, identifiers, commands, events, or public APIs. They do not keep shadow copies that can drift.

For example, Object owns object identity and intrinsic geometry meaning. Placement owns where an object is placed. A renderer may project both into a scene, but renderer objects are not Core state.

Composition Is Additive

Atoms declare `requires` and `provides` tokens. Composition resolves those edges into an ordered plan. Local idempotence makes repeated installation safe; the composition receipt makes the complete accepted plan exactly-once across process restarts.

Core Does Not Mean Default

A capability may be universal and still opt-in. Core means the behavior satisfies the ownership contract. It does not mean every engine instance installs it automatically.

Data Is Not A Runtime Owner

Authored presets, balance values, levels, quests, materials, and complete game loops remain recipe or product data. Core can define the schema and deterministic evaluator while the product owns the authored values.

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Build A First Runtime

Install the package, import only the semantic surfaces you need, construct the Engine, then advance it with explicit time.

Minimal Engine

The runnable source for this example is `examples/guide/basic-engine.mjs`.

```
import { createEngine } from "nexusengine";

const engine = createEngine({ kits: [] });
engine.tick(1 / 60);
console.log(engine.clock.frame);
```

The root API is intentionally small. Domain factories and atoms are imported from generated semantic subpaths.

Add Object And Placement

The runnable source is `examples/guide/object-placement.mjs`.

```
import { createEngine } from "nexusengine";
import { createObjectRegistryKit } from "nexusengine/domains/object/registry";
import { createObjectPlacementKit } from "nexusengine/domains/object/placement";

const engine = createEngine({
  kits: [createObjectRegistryKit(), createObjectPlacementKit()]
});
```

Placement requires the Object descriptor contract. Composition can resolve that provider automatically; direct construction requires you to install both.

Inspect Installed Surfaces

Domain Service Kits register semantic paths and named APIs. Use the engine's addressability APIs instead of reaching into private state.

```
engine.n.path("n:object");
engine.n.path("n:object:placement");
engine.n.ownerOf("n:object");
engine.n.paths();
engine.n.apis();
```

Keep Time Explicit

Pass elapsed time into `tick`. Do not read wall-clock time inside deterministic Core systems. A host may sample a clock, but it converts that sample into explicit runtime input before Core processes it.

Save And Restore

Each stateful atom exposes its manifest-declared snapshot and reset behavior. Applications should persist the complete host composition receipt beside application state so the accepted runtime can be reconstructed after restart.

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Domains And Atomic Kits

The semantic path tells you where a responsibility belongs. The manifest tells tools and humans what that owner may and may not do.

Domain Manifest V2

Every top-level semantic Domain declares:

- stable identity, path, immediate parent, label, version status
- one responsibility and explicit owned meaning
- forbidden responsibilities
- owned state and schema
- inputs, systems, outputs, and lifecycle
- required and optional dependencies
- public atomic Kits, providers, and adapters
- settings schema and proof references

Generation fails when a public atom lacks source or proof. Compliance is never inferred from a folder name.

Semantic Families

Family	Main paths
Runtime	n:runtime , realtime, data, transaction, persistence, sequence, startup
Composition	n:composition , n:mcp , n:policy
World	n:spatial , n:object , n:world , scene, weather, asset
Simulation	n:simulation , physics, motion, n:compute , model
Actors	n:actor , creature, character, player, n:agent
Interaction	n:interaction , input
Presentation	output, graphics, camera, animation, audio, UI, speech, capture, sky
Infrastructure	n:network , n:diagnostics , contract-only n:host

The generated Domain Index appendix contains every active path.

Atomicity Test

An atom should have one reason to change. If a module owns inventory, combat, rendering, and progression together, it is a composition or product feature, not an atom.

Splitting is semantic, not file-count driven. Several source files may implement one indivisible lifecycle. Conversely, one legacy file may contain several owners and must be separated.

Dependency Tokens

Use capability tokens for behavior dependencies and Domain paths for semantic presence. Requirements must describe what the atom consumes; providers list what they supply. Composition rejects missing providers, cycles, identity collisions, and changed content.

Public Imports

Only generated package exports are public. Production code does not import private files from sibling Domains. Cross-Domain collaboration goes through a public subpath or an explicit adapter.

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State, Lifecycle, And Idempotence

Idempotence means repeating an accepted operation does not create a second effect. It is enforced locally by each atom and globally by the composition controller.

Install

An atomic Kit has a stable identity and manifest fingerprint. Installing matching content again returns the installed API without adding state, systems, or subscriptions. Reusing the same identity with different content is a hard conflict.

Commands

State-changing commands need a stable idempotency key when they may be retried. The state owner records the accepted key and result. Repeating the same command returns the original result. Reusing the key with changed content fails before mutation.

Snapshot

A snapshot contains portable semantic state. It excludes renderer objects, functions, sockets, file handles, GPU resources, and vendor SDK instances. Providers snapshot their own implementation state behind their contract when necessary.

Reset

Reset restores the configured baseline. Calling reset twice returns the same snapshot and must not duplicate resources or lifecycle hooks.

Replay

Deterministic replay starts from an accepted snapshot and applies the same ordered inputs. Any nondeterministic source is injected by the host and recorded as input. Random behavior uses explicit seeds or provider receipts.

Composition-Level Exactly-Once

The controller hashes the normalized request, dependency plan, exact Kit source descriptors, configuration, and resolved executable fingerprints. The resulting plan ID is reviewed by a human. After apply, its receipt is persisted. Repeating the same plan returns that receipt without invoking the host again.

Transactional Failure

Before apply, the host supplies a snapshot. If host mutation or receipt persistence fails, the controller restores both host and controller state. A failed plan has no success receipt.

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Composition And Recipes

Composition turns semantic intent into a deterministic, reviewable installation order.

Registry First

Core discovery starts from merged registry metadata generated from manifests. It never scans the filesystem at runtime. Imported registries may add identities but may not replace Core records or collide with an existing Domain path.

Selection Inputs

A request may select:

- individual atomic Kit IDs
- Domain IDs, which select the Domain's public atoms
- recipe IDs, which select declared Domains and Kits
- a scoped composition tree
- per-Kit configuration and allowed statuses

The retired `bundles` request key is rejected. Use recipes for declarative composition.

Validate Before Plan

Validation checks identity, dependencies, provider availability, status policy, cycles, and ordering without resolving executable code.

The runnable inspection example is `examples/guide/composition-inspection.mjs`.

```
const validation = engine.n.composition.planning.validate({
  kits: ["object-placement-kit"]
});

if (!validation.ok) throw new Error("Composition is invalid");
```

Plan

Planning adds required providers and returns a stable dependency order. MCP planning also asks the application host to preflight exact executable sources. Preflight is read-only.

Recipe Ownership

A recipe is data. It can express a useful combination without creating a new state owner. Reusable optional recipes belong in a trusted Kit registry. Complete game recipes belong with the game.

Apply

Apply requires the exact reviewed plan ID and explicit authorization. The controller resolves the same metadata, rejects drift, applies through the host, persists a receipt, and returns the original receipt on replay.

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MCP Agent Workflow

MCP makes the semantic catalog inspectable by an agent. It does not make the agent the runtime owner and does not authorize mutation by itself.

Opt-In Boundary

An application installs the MCP registry Domain, registers the Composition provider, supplies a transactional host, and chooses a transport. Without those steps, the package exposes no live MCP server.

Read Tools

Tool	Purpose
<code>domains_list</code> , <code>domain_get</code>	Discover semantic ownership
<code>kits_list</code> , <code>kit_explain</code>	Preserve the existing Kit-oriented inspection surface
<code>atoms_list</code> , <code>atom_get</code>	Page through atomic capabilities and dependency evidence
<code>recipes_list</code> , <code>recipe_get</code>	Inspect declarative compositions
<code>registry_sources_list</code>	Review package, version, commit, integrity, environment, permissions, and status
<code>composition_validate</code>	Check a request without resolving code
<code>composition_plan</code>	Resolve exact sources and produce a stable plan ID

Mutation Tool

`composition_apply` is the only mutation tool in the provider. It requires explicit authorization for the exact plan ID. A transport or application may impose additional policy.

Review Sequence

```
inspect relevant Domains and atoms
-> validate the request
-> plan exact sources
-> show commit, integrity, permissions, and capability changes
-> human approves the exact plan ID
-> apply once
-> persist receipt
-> runtime continues after MCP disconnects
```

Prompts

The provider exposes exactly two short prompts: `inspect-and-plan` and `review-and-apply`. They route the workflow; they do not embed the full guide.

Resources

Registry records and guide chapters are individual MCP resources. An agent reads only the records and chapters relevant to its current decision. The complete PDF is for humans, not prompt context.

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Registries And Security

A registry record is metadata. Reading, searching, merging, or validating it must not execute package code.

Exact Source Identity

An executable Kit record includes:

- package name and exact version
- canonical package subpath and export name
- immutable 40-character source commit
- SHA-256 integrity
- environments and requested permissions
- requires and provides tokens
- status and settings schema

Moving branches, tags, and unpinned URLs are not executable identities.

Resolution Rules

The host verifies that paths stay inside the resolved package root, imports the declared export only, compares integrity, and computes an executable fingerprint. Browser modules require immutable commit URLs. Local and package paths cannot escape their package root.

No Runtime Installation

Core never runs a package manager during composition. When required code is absent, planning returns a structured installation receipt with the exact package and version. Installation is a separate human-controlled operation; planning is repeated afterward.

Collision Rules

- An imported registry cannot replace a Core identity.
- Two Domains cannot claim the same semantic path.
- One Kit ID cannot refer to changed content.
- A repeated accepted plan returns its original receipt.
- Integrity mismatch, wrong export, path escape, or missing source fails before mutation.

Trust Boundary

Integrity proves identity, not benevolence. Approved JavaScript still runs with the permissions of its host. Review source ownership, commit, permissions, and environment before approval. Use process or platform isolation when untrusted code must run.

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Hosts, Providers, And Adapters

Core owns semantic contracts. Concrete platform work belongs at a leaf boundary.

Host

The host owns application lifecycle, package availability, executable resolution, runtime construction, persistence, and rollback. Core retains only portable host capability contracts under `n:host`.

Browser, Node, native, editor, terminal, repository, filesystem, storage, transport, and vendor SDK implementations are outside Core.

Provider

A provider implements one Domain contract. Examples include physics solving, model inference, speech synthesis, asset retrieval, or capture rendering. Core can own request and result schemas while the provider owns vendor handles, caches, and external effects.

Adapter

An adapter translates two public contracts without becoming the state owner of either. Its manifest names both requirements and the capability it provides. Adapters import public semantic subpaths, never private sibling files.

Presentation Boundary

Presentation Core contains renderer-neutral descriptors and policies. Three.js objects, WebGL resources, browser audio, native widgets, and authored presets belong in renderer, host, Kit, or product repositories.

Example Ownership Decisions

- Generic sky descriptors: `n:presentation:sky`.
- Authored sunset preset: recipe or product data.
- Speech request lifecycle: `n:presentation:speech`.
- PocketTTS implementation: external provider Kit.
- Motion trajectory: `n:simulation:motion`.
- Animation clip descriptor: `n:presentation:animation`.
- Two-bone pose solving: `n:simulation:motion`.

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Testing And Proof

Tests prove contracts at the level where failure would matter. A green unit test is not enough when a change affects package exports, composition, consumers, or rendered output.

Manifest Proof

Validate unique IDs, normalized paths, immediate parents, one owner per public module, source paths, export names, schemas, lifecycle declarations, dependencies, and proof references.

Atomic Lifecycle Proof

For each stateful atom:

1. Install twice and prove no duplicate state or systems.
2. Apply the same command twice and compare state and result.
3. Reuse an idempotency key with changed content and require failure.
4. Snapshot, load, and compare equality.
5. Reset twice and compare equality.
6. Replay ordered inputs from the same snapshot.

Composition Proof

Test missing providers, cycles, collisions, status rejection, stable ordering, repeated apply, process restart, receipt persistence, host rollback, and continued runtime operation after MCP disconnect.

Combination Coverage

Test each atom alone, every declared dependency edge, pairwise cross-Domain combinations, official recipes, high-risk three-way stacks, and installation-order permutations. Do not attempt every mathematical combination.

Registry Security Proof

Prove that metadata reads cannot execute code. Reject moving refs, integrity mismatch, path escape, wrong export, missing package, duplicate identity, and changed-content replay before mutation.

Consumer Proof

Install the packed Engine tarball in clean temporary directories. Then prove Kits, Editor, and Simulator against that exact artifact without local symlinks. Browser-facing workflows require Playwright and human-view inspection in addition to DOM assertions.

Documentation Proof

Regenerate all derived files, run drift checks, execute guide examples, resolve links, render the PDF, verify every page contains content, and visually inspect rendered page images for clipping or overlap.

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Migrating To 0.0.4

0.0.4 is a hard semantic cutover. Old paths and root symbols are removed in the same change as their replacements. There are no runtime forwarding exports.

Import Strategy

Keep root imports for bootstrap and runtime contracts. Import Domain factories and atoms from generated semantic subpaths.

```
import { createEngine } from "nexusengine";
import { createMotionKit } from "nexusengine/domains/simulation/motion";
import { createPhysicsKit } from "nexusengine/domains/simulation/physics";
```

Do not use `n:core-*`, `nexusengine/core-kits/*`, or private source paths.

Runtime Paths

Use semantic paths such as `n:object`, `n:runtime:transaction`, and `n:presentation:graphics`. Immediate parent relationships are validated by manifests.

Removed Implementations

Concrete hosts, renderers, shaders, platform storage, model mocks, speech engines, authored sky presets, gameplay systems, and complete games moved outside Core or became caller-owned data. The Root Migration Map appendix lists every retired source and its owner.

Consumer Migration Order

1. Replace old imports with generated semantic package subpaths.
2. Replace old runtime paths with semantic paths.
3. Install explicit atoms that were formerly implicit defaults.
4. Move platform implementations into the host or an external Kit.
5. Replace compatibility aliases with the owning Domain API.
6. Run from a packed Engine artifact in a clean directory.

Changelog Contract

The changelog and migration map explain replacements. They are not compatibility code. A removed symbol stays removed.

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ProtoKit Extraction

ProtoKits is retired as an active development destination. Its history remains evidence for reconstructing useful behavior with Core atoms, external Kits, recipes, and game-owned code.

Frozen Source

Extraction uses a clean read-only worktree at one immutable remote default-branch commit. The inventory records the remote, commit, file count, and source snapshot SHA-256. A stale or dirty local checkout is never canonical evidence.

Unit Of Classification

Folder count is not capability count. The extractor records folders, files, exports, factories, manifests, tests, dependencies, state, lifecycle signals, commands, events, providers, adapters, and consumers. Every discovered source surface receives exactly one disposition.

Dispositions

Disposition	Meaning
core-reuse	Existing Core atom already owns the behavior
core-composition	Existing atoms reproduce it through composition
core-new-atom	A missing universal atom passed every promotion gate
external-kit	Reusable but optional, niche, platform, or vendor behavior
recipe-data	Authored data or declarative composition
game-owned	Complete product or gameplay behavior
duplicate	Semantically identical to a named canonical owner
rejected-unproven	Insufficient evidence for a safe owner

New Atom Gate

A candidate must have one indivisible responsibility, deterministic behavior or isolated nondeterminism, product neutrality, snapshot and reset support, no concrete presentation ownership, and two semantically different consumers. Failure of one condition keeps it outside Core.

Reconstruction

Useful behavior is preserved as a recipe: source lineage, required Core atoms, external owner, data, policy, and proof status. Preserving behavior does not require preserving the old package shape.

Archive Boundary

Writers and automations remain disabled. Archiving the remote repository is a separate external mutation that requires explicit approval after extraction coverage is complete.

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Release And Operating Model

Release proof is performed against committed, reproducible sources. Local-green evidence is necessary but is not the same as a release.

Generated Truth

The Core catalog, package exports, ownership ledger, API reference, guide indexes, MCP chapter resources, and guide identify the same registry SHA-256. Generate mode writes them; check mode rejects drift.

Required Gates

```
npm ci --ignore-scripts --no-audit --no-fund
npm run core:check
npm run ownership:generate
npm run migrations:check
npm run protokits:check
npm run guide:check
npm run guide:examples
npm test
npm run test:release
npm run boundaries:check
npm run docs:check
npm pack --dry-run --json
```

Consumer repositories must install the exact packed or committed Engine source. Local symlinks and sibling private imports invalidate clean-consumer proof.

Human Approval

After every local and hosted gate passes, present the repository identity, previous numeric branch, derived next version, exact default-branch SHA, checks, and warnings. Approval applies only to that tuple.

Immediately before branch creation, refetch the remote and confirm the default branch still points to the approved SHA. Create the immutable numeric branch with a non-force exact-SHA push, then read the remote ref back and compare it byte for byte.

Outside This Guide Build

This documentation build does not push, publish, archive ProtoKits, mutate Google Drive, create a GitHub Release, publish npm, or create the numeric version branch. Each action has its own approval gate.

GENERATED REFERENCE

Domain Index

Registry SHA-256: fb253d7c33d1b271857591e21f6eaca1f32e470385d6080a131813261c767cc8

- `n:actor` : Own neutral embodied actor identity and shared actor references.
- `n:actor:creature` : Own neutral creature embodiment definitions and references.
- `n:actor:character` : Own active embodied character identity and neutral runtime bindings.
- `n:actor:player` : Own neutral player identity, possession, control authority, and spawn generations.
- `n:agent` : Own product-neutral observation, proposal, decision-cycle, execution receipt, and replay evidence contracts.
- `n:asset` : Own asset identity, manifests, bundles, content-addressed jobs, readiness, and provider contracts.
- `n:composition` : Own deterministic Domain and Kit discovery, dependency planning, plan identity, and exactly-once apply receipts.
- `n:compute` : Own parallel compute descriptors, dependency graphs, dispatch plans, and provider contracts.
- `n:compute:model` : Own model descriptors, registries, inference requests/results, and model provider contracts.
- `n:diagnostics` : Own renderer-neutral telemetry, health, determinism, performance, replay, and debug evidence descriptors.
- `n:host` : Own host capability descriptors and fallback contracts without platform implementation.
- `n:interaction` : Own targets, affordances, activation progress, semantic requirements, prompts, and completion results.
- `n:interaction:input` : Own semantic input actions, axes, contexts, bindings, dead zones, and adapter contracts.
- `n:mcp` : Own opt-in transport-neutral MCP contracts, provider registration, authorization, and protocol dispatch.
- `n:network` : Own session, peer, message, synchronization, authority, latency, reconnect, and collaboration contracts.
- `n:object` : Own renderer-neutral object identity, intrinsic geometry meaning, fidelity, vegetation identity, and placement.
- `n:object:shape` : Own source and derived geometric shapes, provider jobs, qualification, and fallback.
- `n:object:fidelity` : Own valid object forms, fidelity packages, readiness, and contextual adaptation.
- `n:object:vegetation` : Own rooted plant species, instances, lifecycle, and deterministic variation.
- `n:object:vegetation:tree` : Own deterministic tree structure, canopy, growth, and fidelity descriptors.
- `n:object:vegetation:foliage` : Own deterministic foliage structure and descriptors.
- `n:object:vegetation:ecology` : Own deterministic vegetation suitability scoring and species selection.
- `n:object:placement` : Own deterministic placement transforms, grounding, alignment, fit, and validation receipts.
- `n:policy` : Own product-neutral permission, guard, sandbox, and runtime safety decisions.
- `n:presentation` : Own renderer-neutral presentation descriptors and output policy contracts.
- `n:presentation:output` : Own surface, safe-area, viewport, aspect, bar, and render-resolution policy.
- `n:presentation:graphics` : Own renderer-neutral material, lighting, VFX, reflection, quality, batch, and render graph descriptors.
- `n:presentation:camera` : Own renderer-neutral camera targets, modes, smoothing, framing, and occlusion policy.
- `n:presentation:animation` : Own animation, pose, blend, rig, and timeline descriptors.
- `n:presentation:audio` : Own audio cue, music, ambience, mix, volume, and spatial audio descriptors.
- `n:presentation:ui` : Own renderer-neutral HUD, menu, prompt, notification, focus, selection, accessibility, and scale descriptors.
- `n:presentation:speech` : Own provider-neutral speech requests, voices, utterance lifecycle, and synthesis result contracts.
- `n:presentation:capture` : Own observation requests, view sets, framing, capture jobs, progress, and result contracts.
- `n:presentation:sky` : Own generic sky, atmosphere, cloud, horizon, and celestial descriptors.
- `n:runtime` : Own deterministic engine lifecycle, ticks, state mutation contracts, and runtime service installation.
- `n:runtime:realtime` : Own deterministic frame context and realtime phase execution.
- `n:runtime:data` : Own schemas, snapshots, selectors, migrations, deterministic random streams, and portable data envelopes.

- `n:runtime:transaction` : Own portable repeat-safe operation and transaction receipts.
- `n:runtime:persistence` : Own save/load targets, save slots, recovery records, and adapter contracts.
- `n:runtime:sequence` : Own deterministic sequence nodes, ordered execution, and frame-driven sequence state.
- `n:runtime:startup` : Own launch truth, preparation facts, continuation choice, structured failure, and readiness receipts.
- `n:simulation` : Own deterministic simulation objectives, resources, hazards, pressure, checkpoints, timers, and resolution contracts.
- `n:simulation:physics` : Own backend-neutral physical bodies, colliders, contacts, constraints, queries, and provider contracts.
- `n:simulation:physics:articulated` : Own articulated body topology, joint dynamics inputs, and backend-neutral articulation state.
- `n:simulation:motion` : Own intent-to-motion descriptors, movement modes, trajectories, velocity state, movement policies, and deterministic pose solving.
- `n:simulation:motion:articulated` : Own target poses, joint limits, articulation motion plans, and drive requests.
- `n:spatial` : Own renderer-neutral transforms, coordinate spaces, bounds, zones, distance queries, and deterministic spatial math.
- `n:world` : Own world identity, cells, partitions, surfaces, deterministic assembly, and world state receipts.
- `n:world:scene` : Own host-neutral scene identity, lifecycle, transition, binding descriptors, and scene snapshots.
- `n:world:weather` : Own weather conditions, tendencies, regions, layers, sampling, and deterministic evolution.
- `n:world:foundation` : Own deterministic foundation contributions, composition, sampling, and cell resolution.
- `n:world:feature` : Own semantic world feature definitions, registries, lifecycle, queries, and composition.
- `n:world:feature:landform` : Own semantic elevation and landform feature descriptors.
- `n:world:feature:hydrology` : Own semantic watershed, water path, water body, and wetland feature descriptors.
- `n:world:feature:ecology` : Own semantic biome, habitat, vegetation-region, and ecotone feature descriptors.
- `n:world:feature:settlement` : Own semantic settlement, route, structure, and infrastructure feature descriptors.
- `n:world:feature:atmosphere` : Own semantic cloud, fog, wind, thermal, precipitation, and visibility feature descriptors.

GENERATED REFERENCE

Core Dependency Table

Registry SHA-256: fb253d7c33d1b271857591e21f6eaca1f32e470385d6080a131813261c767cc8

Owner	Requires	Optional
n:actor	-	-
n:actor:creature	n:actor	-
n:actor:character	n:actor	-
n:actor:player	n:actor:character	-
n:agent	-	-
n:asset	-	-
n:composition	-	n:mcp
n:compute	-	-
n:compute:model	n:compute	-
n:diagnostics	-	-
n:host	-	-
n:interaction	-	-
n:interaction:input	-	-
n:mcp	-	n:composition
n:network	-	-
n:object	-	n:asset , n:simulation:physics
n:object:shape	object:descriptor-contract	-
n:object:fidelity	object:descriptor-contract	-
n:object:vegetation	n:object	-
n:object:vegetation:tree	n:object:vegetation	-
n:object:vegetation:foliage	n:object:vegetation	-
n:object:vegetation:ecology	n:object:vegetation	-
n:object:placement	object:descriptor-contract	-
n:policy	-	-
n:presentation	-	-
n:presentation:output	n:presentation	-
n:presentation:graphics	n:presentation	-
n:presentation:camera	n:presentation	-
n:presentation:animation	n:presentation	-
n:presentation:audio	n:presentation	-
n:presentation:ui	n:presentation	-
n:presentation:speech	n:presentation	-
n:presentation:capture	n:presentation	-

Owner	Requires	Optional
n:presentation:sky	n:presentation	-
n:runtime	-	-
n:runtime:realtime	n:runtime	-
n:runtime:data	n:runtime	-
n:runtime:transaction	n:runtime	-
n:runtime:persistence	n:runtime:data	-
n:runtime:sequence	n:runtime	-
n:runtime:startup	n:runtime	n:asset
n:simulation	n:runtime:realtime	-
n:simulation:physics	n:simulation	-
n:simulation:physics:articulated	n:simulation:physics	-
n:simulation:motion	n:simulation	-
n:simulation:motion:articulated	n:simulation:motion	-
n:spatial	-	-
n:world	n:spatial	-
n:world:scene	n:world	-
n:world:weather	n:world	-
n:world:foundation	n:world	-
n:world:feature	n:world	-
n:world:feature:landform	n:world:feature	-
n:world:feature:hydrology	n:world:feature	-
n:world:feature:ecology	n:world:feature	-
n:world:feature:settlement	n:world:feature	-
n:world:feature:atmosphere	n:world:feature	-

GENERATED REFERENCE

Generated Core API Reference

This file is generated from Domain manifest v2 records. Do not edit it directly.

Registry SHA-256: fb253d7c33d1b271857591e21f6eaca1f32e470385d6080a131813261c767cc8

Domains

Domain	Parent	Responsibility	Status
n:actor	-	Own neutral embodied actor identity and shared actor references.	stable-candidate
n:actor:creature	n:actor	Own neutral creature embodiment definitions and references.	stable-candidate
n:actor:character	n:actor	Own active embodied character identity and neutral runtime bindings.	stable-candidate
n:actor:player	n:actor	Own neutral player identity, possession, control authority, and spawn generations.	stable-candidate
n:agent	-	Own product-neutral observation, proposal, decision-cycle, execution receipt, and replay evidence contracts.	stable-candidate
n:asset	-	Own asset identity, manifests, bundles, content-addressed jobs, readiness, and provider contracts.	stable-candidate
n:composition	-	Own deterministic Domain and Kit discovery, dependency planning, plan identity, and exactly-once apply receipts.	stable-candidate
n:compute	-	Own parallel compute descriptors, dependency graphs, dispatch plans, and provider contracts.	stable-candidate
n:compute:model	n:compute	Own model descriptors, registries, inference requests/results, and model provider contracts.	stable-candidate
n:diagnostics	-	Own renderer-neutral telemetry, health, determinism, performance, replay, and debug evidence descriptors.	stable-candidate
n:host	-	Own host capability descriptors and fallback contracts without platform implementation.	stable-candidate
n:interaction	-	Own targets, affordances, activation progress, semantic requirements, prompts, and completion results.	stable-candidate
n:interaction:input	n:interaction	Own semantic input actions, axes, contexts, bindings, dead zones, and adapter contracts.	stable-candidate

Domain	Parent	Responsibility	Status
n:mcp	-	Own opt-in transport-neutral MCP contracts, provider registration, authorization, and protocol dispatch.	stable-candidate
n:network	-	Own session, peer, message, synchronization, authority, latency, reconnect, and collaboration contracts.	stable-candidate
n:object	-	Own renderer-neutral object identity, intrinsic geometry meaning, fidelity, vegetation identity, and placement.	stable-candidate
n:object:shape	n:object	Own source and derived geometric shapes, provider jobs, qualification, and fallback.	stable-candidate
n:object:fidelity	n:object	Own valid object forms, fidelity packages, readiness, and contextual adaptation.	stable-candidate
n:object:vegetation	n:object	Own rooted plant species, instances, lifecycle, and deterministic variation.	stable-candidate
n:object:vegetation:tree	n:object:vegetation	Own deterministic tree structure, canopy, growth, and fidelity descriptors.	stable-candidate
n:object:vegetation:foliage	n:object:vegetation	Own deterministic foliage structure and descriptors.	stable-candidate
n:object:vegetation:ecology	n:object:vegetation	Own deterministic vegetation suitability scoring and species selection.	stable-candidate
n:object:placement	n:object	Own deterministic placement transforms, grounding, alignment, fit, and validation receipts.	stable-candidate
n:policy	-	Own product-neutral permission, guard, sandbox, and runtime safety decisions.	stable-candidate
n:presentation	-	Own renderer-neutral presentation descriptors and output policy contracts.	stable-candidate
n:presentation:output	n:presentation	Own surface, safe-area, viewport, aspect, bar, and render-resolution policy.	stable-candidate
n:presentation:graphics	n:presentation	Own renderer-neutral material, lighting, VFX, reflection, quality, batch, and render graph descriptors.	stable-candidate
n:presentation:camera	n:presentation	Own renderer-neutral camera targets, modes, smoothing, framing, and occlusion policy.	stable-candidate
n:presentation:animation	n:presentation	Own animation, pose, blend, rig, and timeline descriptors.	stable-candidate

Domain	Parent	Responsibility	Status
n:presentation:audio	n:presentation	Own audio cue, music, ambience, mix, volume, and spatial audio descriptors.	stable-candidate
n:presentation:ui	n:presentation	Own renderer-neutral HUD, menu, prompt, notification, focus, selection, accessibility, and scale descriptors.	stable-candidate
n:presentation:speech	n:presentation	Own provider-neutral speech requests, voices, utterance lifecycle, and synthesis result contracts.	stable-candidate
n:presentation:capture	n:presentation	Own observation requests, view sets, framing, capture jobs, progress, and result contracts.	stable-candidate
n:presentation:sky	n:presentation	Own generic sky, atmosphere, cloud, horizon, and celestial descriptors.	stable-candidate
n:runtime	-	Own deterministic engine lifecycle, ticks, state mutation contracts, and runtime service installation.	stable-candidate
n:runtime:realtime	n:runtime	Own deterministic frame context and realtime phase execution.	stable-candidate
n:runtime:data	n:runtime	Own schemas, snapshots, selectors, migrations, deterministic random streams, and portable data envelopes.	stable-candidate
n:runtime:transaction	n:runtime	Own portable repeat-safe operation and transaction receipts.	stable-candidate
n:runtime:persistence	n:runtime	Own save/load targets, save slots, recovery records, and adapter contracts.	stable-candidate
n:runtime:sequence	n:runtime	Own deterministic sequence nodes, ordered execution, and frame-driven sequence state.	stable-candidate
n:runtime:startup	n:runtime	Own launch truth, preparation facts, continuation choice, structured failure, and readiness receipts.	stable-candidate
n:simulation	-	Own deterministic simulation objectives, resources, hazards, pressure, checkpoints, timers, and resolution contracts.	stable-candidate
n:simulation:physics	n:simulation	Own backend-neutral physical bodies, colliders, contacts, constraints, queries, and provider contracts.	stable-candidate
n:simulation:physics:articulated	n:simulation:physics	Own articulated body topology, joint dynamics inputs, and backend-neutral articulation state.	stable-candidate
n:simulation:motion	n:simulation	Own intent-to-motion descriptors, movement modes, trajectories, velocity state, movement policies, and deterministic pose solving.	stable-candidate

Domain	Parent	Responsibility	Status
n:simulation:motion:articulated	n:simulation:motion	Own target poses, joint limits, articulation motion plans, and drive requests.	stable-candidate
n:spatial	-	Own renderer-neutral transforms, coordinate spaces, bounds, zones, distance queries, and deterministic spatial math.	stable-candidate
n:world	-	Own world identity, cells, partitions, surfaces, deterministic assembly, and world state receipts.	stable-candidate
n:world:scene	n:world	Own host-neutral scene identity, lifecycle, transition, binding descriptors, and scene snapshots.	stable-candidate
n:world:weather	n:world	Own weather conditions, tendencies, regions, layers, sampling, and deterministic evolution.	stable-candidate
n:world:foundation	n:world	Own deterministic foundation contributions, composition, sampling, and cell resolution.	stable-candidate
n:world:feature	n:world	Own semantic world feature definitions, registries, lifecycle, queries, and composition.	stable-candidate
n:world:feature:landform	n:world:feature	Own semantic elevation and landform feature descriptors.	stable-candidate
n:world:feature:hydrology	n:world:feature	Own semantic watershed, water path, water body, and wetland feature descriptors.	stable-candidate
n:world:feature:ecology	n:world:feature	Own semantic biome, habitat, vegetation-region, and ecotone feature descriptors.	stable-candidate
n:world:feature:settlement	n:world:feature	Own semantic settlement, route, structure, and infrastructure feature descriptors.	stable-candidate
n:world:feature:atmosphere	n:world:feature	Own semantic cloud, fog, wind, thermal, precipitation, and visibility feature descriptors.	stable-candidate

Atomic Kits

Kit	Domain	Public subpath	Responsibility
actor-registry-kit	n:actor	nexusengine/domains/actor/registry	Own neutral actor identities and embodiment references.
creature-registry-kit	n:actor:creature	nexusengine/domains/actor/creature	Register and resolve neutral creature embodiment definitions.
character-registry-kit	n:actor:character	nexusengine/domains/actor/character	Register and resolve active embodied character descriptors.
player-authority-kit	n:actor:player	nexusengine/domains/actor/player	Track player identity, possession, control authority, and spawn generations.

Kit	Domain	Public subpath	Responsibility
agent-cycle-kit	n:agent	nexusengine/domains/agent/cycle	Record observations, action proposals, decision cycles, and execution receipts.
asset-registry-kit	n:asset	nexusengine/domains/asset/registry	Resolve asset manifests and bundles through content-addressed provider jobs.
composition-registry-kit	n:composition	nexusengine/domains/composition/registry	Maintain normalized composition metadata and produce deterministic plans and receipts.
compute-graph-kit	n:compute	nexusengine/domains/compute/graph	Validate compute descriptors and create deterministic dependency-ordered dispatch plans.
model-registry-kit	n:compute:model	nexusengine/domains/compute/model	Register model descriptors and normalize provider-neutral inference requests and results.
diagnostics-kit	n:diagnostics	nexusengine/domains/diagnostics/runtime	Collect serializable telemetry, runtime health, determinism, and performance evidence.
debug-descriptor-kit	n:diagnostics	nexusengine/domains/diagnostics/debug	Record renderer-neutral rays, markers, scalars, and capture packets for diagnostics.
debug-draw-descriptor-kit	n:diagnostics	nexusengine/domains/diagnostics/debug-draw	Create stateless renderer-neutral debug draw descriptors.
host-capability-kit	n:host	nexusengine/domains/host/capabilities	Describe available host capabilities and select declarative fallback modes.
interaction-kit	n:interaction	nexusengine/domains/interaction/runtime	Manage interaction targets, affordances, activation, and results.
input-contract-kit	n:interaction:input	nexusengine/domains/interaction/input	Normalize semantic input actions, axes, contexts, and bindings.
mcp-registry-kit	n:mcp	nexusengine/domains/mcp/registry	Register and dispatch schema-valid MCP providers through an explicit authorization boundary.
network-contract-kit	n:network	nexusengine/domains/network/contracts	Describe network sessions, messages, authority, and synchronization without owning transport.
object-registry-kit	n:object	nexusengine/domains/object/registry	Own object identity and renderer-neutral lifecycle records.
object-shape-kit	n:object:shape	nexusengine/domains/object/shape	Derive and qualify renderer-neutral geometric shape candidates.
object-fidelity-kit	n:object:fidelity	nexusengine/domains/object/fidelity	Package and select valid object fidelity forms.
object-vegetation-kit	n:object:vegetation	nexusengine/domains/object/vegetation	Own deterministic plant species, instances, and lifecycle state.
object-tree-kit	n:object:vegetation:tree	nexusengine/domains/object/vegetation/tree	Produce deterministic tree structure, canopy, growth, and fidelity descriptors.
object-foliage-kit	n:object:vegetation:foliage	nexusengine/domains/object/vegetation/foliage	Produce deterministic foliage structures and descriptors.

Kit	Domain	Public subpath	Responsibility
object-vegetation-ecology-kit	n:object:vegetation:ecology	nexusengine/domains/object/vegetation/ecology	Score vegetation suitability and select species deterministically.
object-placement-kit	n:object:placement	nexusengine/domains/object/placement	Create, validate, and replay deterministic object placement receipts.
object-meshoptimizer-shape-provider-kit	n:object:shape	nexusengine/domains/object/shape/meshoptimizer-provider	Resolve object shape jobs through an explicitly registered meshoptimizer-compatible provider.
object-shape-fidelity-adapter-kit	n:object:fidelity	nexusengine/domains/object/adapters/shape-fidelity	Translate qualified shape records into Object Fidelity form requests.
object-vegetation-bridge-kit	n:object:vegetation	nexusengine/domains/object/adapters/vegetation-object	Project vegetation identities into canonical Object descriptors without owning either state.
policy-kit	n:policy	nexusengine/domains/policy/guard	Evaluate declarative runtime safety and permission rules.
presentation-registry-kit	n:presentation	nexusengine/domains/presentation/registry	Register renderer-neutral presentation capabilities and descriptors.
presentation-output-kit	n:presentation:output	nexusengine/domains/presentation/output	Calculate renderer-neutral surfaces, safe areas, viewports, aspect policy, and render resolution.
graphics-descriptor-kit	n:presentation:graphics	nexusengine/domains/presentation/graphics	Create renderer-neutral material, lighting, VFX, quality, batch, and terrain LOD descriptors.
render-layer-graph-kit	n:presentation:graphics	nexusengine/domains/presentation/graphics/render-graph	Validate renderer-neutral render layers, dependencies, and pass ordering.
reflection-descriptor-kit	n:presentation:graphics	nexusengine/domains/presentation/graphics/reflection	Describe renderer-neutral reflection probes, plans, and update policy.
camera-descriptor-kit	n:presentation:camera	nexusengine/domains/presentation/camera	Manage camera targets, modes, smoothing, and renderer-neutral camera policy.
camera-framing-kit	n:presentation:camera	nexusengine/domains/presentation/camera/framing	Calculate perspective and orthographic subject framing for a viewport.
camera-control-math-kit	n:presentation:camera	nexusengine/domains/presentation/camera/control-math	Calculate camera-relative yaw, orbit limits, and root handoff state.
animation-descriptor-kit	n:presentation:animation	nexusengine/domains/presentation/animation	Manage animation clips, poses, blends, transitions, and timeline descriptors.
rig-transform-kit	n:presentation:animation	nexusengine/domains/presentation/animation/rig	Create neutral joint, limb, and rig transform descriptors.
audio-descriptor-kit	n:presentation:audio	nexusengine/domains/presentation/audio	Manage renderer-neutral audio cues, mix groups, volume policy, and spatial audio descriptors.
ui-descriptor-kit	n:presentation:ui	nexusengine/domains/presentation/ui	Manage renderer-neutral UI, focus, selection, prompt, notification, and accessibility descriptors.

Kit	Domain	Public subpath	Responsibility
ui-scale-kit	n:presentation:ui	nexusengine/domains/presentation/ui/scale	Calculate deterministic reference-resolution and UI scale policy.
speech-contract-kit	n:presentation:speech	nexusengine/domains/presentation/speech	Manage provider-neutral speech requests, voices, utterance lifecycle, and synthesis results.
capture-contract-kit	n:presentation:capture	nexusengine/domains/presentation/capture	Manage observation requests, view sets, framing, capture jobs, progress, and result contracts.
sky-descriptor-kit	n:presentation:sky	nexusengine/domains/presentation/sky	Create generic sky, horizon, atmosphere, cloud, and celestial descriptors.
runtime-lifecycle-kit	n:runtime	nexusengine/domains/runtime/lifecycle	Own deterministic runtime lifecycle and Kit installation receipts.
realtime-runtime-kit	n:runtime:realtime	nexusengine/domains/runtime/realtime	Create deterministic realtime frame context and phase execution.
runtime-data-kit	n:runtime:data	nexusengine/domains/runtime/data	Provide deterministic schemas, snapshots, selectors, migrations, and data envelopes.
transaction-ledger-kit	n:runtime:transaction	nexusengine/domains/runtime/transaction	Record repeat-safe operation keys and immutable transaction receipts.
persistence-contract-kit	n:runtime:persistence	nexusengine/domains/runtime/persistence	Describe save/load targets, slots, recovery records, and persistence adapter contracts.
runtime-sequence-kit	n:runtime:sequence	nexusengine/domains/runtime/sequence	Install deterministic sequence node definitions and execution state.
runtime-startup-kit	n:runtime:startup	nexusengine/domains/runtime/startup	Coordinate deterministic startup preparation and readiness receipts.
simulation-state-kit	n:simulation	nexusengine/domains/simulation/runtime	Manage deterministic simulation objectives, resources, hazards, timers, and resolution receipts.
physics-contract-kit	n:simulation:physics	nexusengine/domains/simulation/physics	Describe physical bodies, colliders, contacts, constraints, queries, and provider boundaries.
articulated-physics-kit	n:simulation:physics:articulated	nexusengine/domains/simulation/physics/articulated	Manage backend-neutral articulated body topology and joint dynamics state.
motion-contract-kit	n:simulation:motion	nexusengine/domains/simulation/motion	Manage intent-to-motion descriptors, trajectories, velocity state, and movement policies.
two-bone-ik-kit	n:simulation:motion	nexusengine/domains/simulation/motion/two-bone-ik	Solve deterministic two-bone inverse-kinematics poses.
articulated-motion-kit	n:simulation:motion:articulated	nexusengine/domains/simulation/motion/articulated	Create target poses, joint limits, articulation plans, and drive requests.
articulated-motion-drive-adapter-kit	n:simulation:physics:articulated	nexusengine/domains/simulation/adapters/articulated-drive	Translate articulated motion plans into backend-neutral physics drive requests.

Kit	Domain	Public subpath	Responsibility
spatial-contract-kit	n:spatial	nexusengine/domains/spatial/contracts	Describe transforms, bounds, zones, spaces, and spatial query requests.
spatial-angle-math-kit	n:spatial	nexusengine/domains/spatial/angle-math	Normalize, compare, and interpolate angular values.
spatial-vector-math-kit	n:spatial	nexusengine/domains/spatial/vector-math	Create and operate on renderer-neutral vector values.
spatial-transform-math-kit	n:spatial	nexusengine/domains/spatial/transform-math	Calculate deterministic transforms, bases, interpolation, and planar projections.
spatial-quaternion-math-kit	n:spatial	nexusengine/domains/spatial/quaternion-math	Create, compose, normalize, rotate, and interpolate quaternions.
world-state-kit	n:world	nexusengine/domains/world/runtime	Manage world identity, cells, partitions, surfaces, and deterministic assembly state.
scene-lifecycle-kit	n:world:scene	nexusengine/domains/world/scene	Manage host-neutral scene identity, lifecycle, transitions, bindings, and snapshots.
weather-state-kit	n:world:weather	nexusengine/domains/world/weather	Manage deterministic weather conditions, tendencies, regions, and sampling.
layered-weather-kit	n:world:weather	nexusengine/domains/world/weather/layers	Compose and evolve deterministic altitude-aware weather layers.
world-foundation-kit	n:world:foundation	nexusengine/domains/world/foundation	Compose deterministic world foundation definitions, contributions, sampling, and cell resolution.
world-feature-kit	n:world:feature	nexusengine/domains/world/feature	Manage semantic world feature definitions, registry, lifecycle, queries, and composition.
semantic-world-feature-kit	n:world:feature	nexusengine/domains/world/feature/semantic	Create one semantic world feature Domain Service Kit from a bounded feature specification.
landform-feature-kit	n:world:feature:landform	nexusengine/domains/world/feature/landform	Create semantic elevation and landform feature descriptors and lifecycle state.
hydrology-feature-kit	n:world:feature:hydrology	nexusengine/domains/world/feature/hydrology	Create semantic watershed, water path, water body, and wetland feature descriptors.
ecology-feature-kit	n:world:feature:ecology	nexusengine/domains/world/feature/ecology	Create semantic biome, habitat, vegetation-region, and ecotone feature descriptors.
settlement-feature-kit	n:world:feature:settlement	nexusengine/domains/world/feature/settlement	Create semantic settlement, route, structure, and infrastructure feature descriptors.
atmosphere-feature-kit	n:world:feature:atmosphere	nexusengine/domains/world/feature/atmosphere	Create semantic cloud, fog, wind, thermal, precipitation, and visibility feature descriptors.

GENERATED REFERENCE

Kit Ownership Ledger

Generated from Domain manifest v2 and the production source inventory. Null compliance fields are intentionally unproven; they are never inferred as true.

Registry SHA-256: fb253d7c33d1b271857591e21f6eaca1f32e470385d6080a131813261c767cc8

- Source modules: 306
- Manifest-proven public atoms: 76
- Manifest-owned internal modules: 205
- Root contract modules: 25
- Unreviewed modules: 0
- Violations: 0

Path	Owner	Review	Destination
src/core-domains/actor/domain.manifest.js	n:actor	manifest-owned-internal	NexusEngine Core
src/core-domains/actor/index.js	n:actor	manifest-owned-internal	NexusEngine Core
src/core-domains/actor/kits/actor-registry-kit/index.js	n:actor	manifest-proven-public-atom	NexusEngine Core
src/core-domains/actor/subdomains/character/index.js	n:actor	manifest-owned-internal	NexusEngine Core
src/core-domains/actor/subdomains/character/kits/character-kit/contracts.js	n:actor	manifest-owned-internal	NexusEngine Core
src/core-domains/actor/subdomains/character/kits/character-kit/index.js	n:actor	manifest-proven-public-atom	NexusEngine Core
src/core-domains/actor/subdomains/creature/index.js	n:actor	manifest-owned-internal	NexusEngine Core
src/core-domains/actor/subdomains/creature/kits/creature-kit/contracts.js	n:actor	manifest-owned-internal	NexusEngine Core
src/core-domains/actor/subdomains/creature/kits/creature-kit/index.js	n:actor	manifest-proven-public-atom	NexusEngine Core
src/core-domains/actor/subdomains/player/index.js	n:actor	manifest-owned-internal	NexusEngine Core
src/core-domains/actor/subdomains/player/kits/player-kit/contracts.js	n:actor	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
src/core-domains/actor/subdomains/player/kits/player-kit/index.js	n:actor	manifest-proven-public-atom	NexusEngine Core
src/core-domains/agent/domain.manifest.js	n:agent	manifest-owned-internal	NexusEngine Core
src/core-domains/agent/index.js	n:agent	manifest-owned-internal	NexusEngine Core
src/core-domains/agent/kits/agent-kit/action-proposals.js	n:agent	manifest-owned-internal	NexusEngine Core
src/core-domains/agent/kits/agent-kit/adapters.js	n:agent	manifest-owned-internal	NexusEngine Core
src/core-domains/agent/kits/agent-kit/agent-state.js	n:agent	manifest-owned-internal	NexusEngine Core
src/core-domains/agent/kits/agent-kit/decision-cycle.js	n:agent	manifest-owned-internal	NexusEngine Core
src/core-domains/agent/kits/agent-kit/execution-ledger.js	n:agent	manifest-owned-internal	NexusEngine Core
src/core-domains/agent/kits/agent-kit/index.js	n:agent	manifest-proven-public-atom	NexusEngine Core
src/core-domains/agent/kits/agent-kit/observations.js	n:agent	manifest-owned-internal	NexusEngine Core
src/core-domains/asset/domain.manifest.js	n:asset	manifest-owned-internal	NexusEngine Core
src/core-domains/asset/index.js	n:asset	manifest-owned-internal	NexusEngine Core
src/core-domains/asset/kits/asset-kit/cache-provider.js	n:asset	manifest-owned-internal	NexusEngine Core
src/core-domains/asset/kits/asset-kit/descriptors.js	n:asset	manifest-owned-internal	NexusEngine Core
src/core-domains/asset/kits/asset-kit/index.js	n:asset	manifest-proven-public-atom	NexusEngine Core
src/core-domains/asset/kits/asset-kit/provider.js	n:asset	manifest-owned-internal	NexusEngine Core
src/core-domains/catalog.js	n:composition	manifest-infrastructure	NexusEngine Core
src/core-domains/composition/adapters/mcp/composition-mcp-provider.js	n:composition	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
src/core-domains/composition/adapters/mcp/generated-guide-resources.js	n:composition	manifest-owned-internal	NexusEngine Core
src/core-domains/composition/domain.manifest.js	n:composition	manifest-owned-internal	NexusEngine Core
src/core-domains/composition/index.js	n:composition	manifest-owned-internal	NexusEngine Core
src/core-domains/composition/kits/composition-registry-kit/composition-tree.js	n:composition	manifest-owned-internal	NexusEngine Core
src/core-domains/composition/kits/composition-registry-kit/index.js	n:composition	manifest-proven-public-atom	NexusEngine Core
src/core-domains/composition/kits/composition-registry-kit/registry.js	n:composition	manifest-owned-internal	NexusEngine Core
src/core-domains/composition/kits/composition-registry-kit/services.js	n:composition	manifest-owned-internal	NexusEngine Core
src/core-domains/composition/services/composition-apply-controller.js	n:composition	manifest-owned-internal	NexusEngine Core
src/core-domains/compute/domain.manifest.js	n:compute	manifest-owned-internal	NexusEngine Core
src/core-domains/compute/index.js	n:compute	manifest-owned-internal	NexusEngine Core
src/core-domains/compute/kits/compute-kit/descriptors.js	n:compute	manifest-owned-internal	NexusEngine Core
src/core-domains/compute/kits/compute-kit/index.js	n:compute	manifest-proven-public-atom	NexusEngine Core
src/core-domains/compute/kits/compute-kit/provider.js	n:compute	manifest-owned-internal	NexusEngine Core
src/core-domains/compute/subdomains/model/kits/model-kit/adapters.js	n:compute	manifest-owned-internal	NexusEngine Core
src/core-domains/compute/subdomains/model/kits/model-kit/index.js	n:compute	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
src/core-domains/compute/subdomains/model/kits/model-kit/inference-request.js	n:compute	manifest-owned-internal	NexusEngine Core
src/core-domains/compute/subdomains/model/kits/model-kit/inference-result.js	n:compute	manifest-owned-internal	NexusEngine Core
src/core-domains/compute/subdomains/model/kits/model-kit/model-descriptors.js	n:compute	manifest-owned-internal	NexusEngine Core
src/core-domains/compute/subdomains/model/kits/model-kit/model-registry.js	n:compute	manifest-owned-internal	NexusEngine Core
src/core-domains/diagnostics/domain.manifest.js	n:diagnostics	manifest-owned-internal	NexusEngine Core
src/core-domains/diagnostics/index.js	n:diagnostics	manifest-owned-internal	NexusEngine Core
src/core-domains/diagnostics/kits/debug-descriptor-kit/index.js	n:diagnostics	manifest-proven-public-atom	NexusEngine Core
src/core-domains/diagnostics/kits/debug-draw-kit.js	n:diagnostics	manifest-proven-public-atom	NexusEngine Core
src/core-domains/diagnostics/kits/diagnostics-kit/index.js	n:diagnostics	manifest-proven-public-atom	NexusEngine Core
src/core-domains/domain-kit.js	n:composition	manifest-infrastructure	NexusEngine Core
src/core-domains/domain-manifest.js	n:composition	manifest-infrastructure	NexusEngine Core
src/core-domains/host/domain.manifest.js	n:host	manifest-owned-internal	NexusEngine Core
src/core-domains/host/index.js	n:host	manifest-owned-internal	NexusEngine Core
src/core-domains/host/kits/host-capability-kit/index.js	n:host	manifest-proven-public-atom	NexusEngine Core
src/core-domains/index.js	n:composition	manifest-infrastructure	NexusEngine Core
src/core-domains/interaction/domain.manifest.js	n:interaction	manifest-owned-internal	NexusEngine Core
src/core-domains/interaction/index.js	n:interaction	manifest-owned-internal	NexusEngine Core
src/core-domains/interaction/kits/interaction-kit/activation.js	n:interaction	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
src/core-domains/interaction/kits/interaction-kit/affordances.js	n:interaction	manifest-owned-internal	NexusEngine Core
src/core-domains/interaction/kits/interaction-kit/index.js	n:interaction	manifest-proven-public-atom	NexusEngine Core
src/core-domains/interaction/kits/interaction-kit/prompts.js	n:interaction	manifest-owned-internal	NexusEngine Core
src/core-domains/interaction/kits/interaction-kit/results.js	n:interaction	manifest-owned-internal	NexusEngine Core
src/core-domains/interaction/kits/interaction-kit/targets.js	n:interaction	manifest-owned-internal	NexusEngine Core
src/core-domains/interaction/subdomains/input/kits/input-kit/actions.js	n:interaction	manifest-owned-internal	NexusEngine Core
src/core-domains/interaction/subdomains/input/kits/input-kit/adapters.js	n:interaction	manifest-owned-internal	NexusEngine Core
src/core-domains/interaction/subdomains/input/kits/input-kit/bindings.js	n:interaction	manifest-owned-internal	NexusEngine Core
src/core-domains/interaction/subdomains/input/kits/input-kit/contexts.js	n:interaction	manifest-owned-internal	NexusEngine Core
src/core-domains/interaction/subdomains/input/kits/input-kit/index.js	n:interaction	manifest-proven-public-atom	NexusEngine Core
src/core-domains/interaction/subdomains/input/kits/input-kit/intent.js	n:interaction	manifest-owned-internal	NexusEngine Core
src/core-domains/manifest-domain-service-kit.js	n:composition	manifest-infrastructure	NexusEngine Core
src/core-domains/manifest-input.js	n:composition	manifest-infrastructure	NexusEngine Core
src/core-domains/manifest-kit-contract.js	n:composition	manifest-infrastructure	NexusEngine Core
src/core-domains/mcp/adapters/node-mcp-sdk-adapter/index.js	n:mcp	manifest-owned-internal	NexusEngine Core
src/core-domains/mcp/contracts/contract-utilities.js	n:mcp	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
src/core-domains/mcp/contracts/prompt-contract.js	n:mcp	manifest-owned-internal	NexusEngine Core
src/core-domains/mcp/contracts/provider-contract.js	n:mcp	manifest-owned-internal	NexusEngine Core
src/core-domains/mcp/contracts/resource-contract.js	n:mcp	manifest-owned-internal	NexusEngine Core
src/core-domains/mcp/contracts/result-contract.js	n:mcp	manifest-owned-internal	NexusEngine Core
src/core-domains/mcp/contracts/tool-contract.js	n:mcp	manifest-owned-internal	NexusEngine Core
src/core-domains/mcp/domain.manifest.js	n:mcp	manifest-owned-internal	NexusEngine Core
src/core-domains/mcp/index.js	n:mcp	manifest-owned-internal	NexusEngine Core
src/core-domains/mcp/kits/mcp-registry-kit/authorization.js	n:mcp	manifest-owned-internal	NexusEngine Core
src/core-domains/mcp/kits/mcp-registry-kit/index.js	n:mcp	manifest-proven-public-atom	NexusEngine Core
src/core-domains/mcp/kits/mcp-registry-kit/registry.js	n:mcp	manifest-owned-internal	NexusEngine Core
src/core-domains/mcp/state/registry-snapshot.js	n:mcp	manifest-owned-internal	NexusEngine Core
src/core-domains/network/domain.manifest.js	n:network	manifest-owned-internal	NexusEngine Core
src/core-domains/network/index.js	n:network	manifest-owned-internal	NexusEngine Core
src/core-domains/network/kits/network-kit/index.js	n:network	manifest-proven-public-atom	NexusEngine Core
src/core-domains/object/adapters/object-shape-fidelity-adapter-kit/index.js	n:object	manifest-proven-public-atom	NexusEngine Core
src/core-domains/object/contracts/object-descriptor.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/domain.manifest.js	n:object	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
src/core-domains/object/index.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/kits/object-registry-kit/index.js	n:object	manifest-proven-public-atom	NexusEngine Core
src/core-domains/object/state/object-registry-state.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/subdomains/fidelity/index.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/subdomains/fidelity/kits/object-fidelity-kit/descriptors.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/subdomains/fidelity/kits/object-fidelity-kit/index.js	n:object	manifest-proven-public-atom	NexusEngine Core
src/core-domains/object/subdomains/placement/contracts/placement-descriptor.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/subdomains/placement/index.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/subdomains/placement/kits/object-placement-kit/index.js	n:object	manifest-proven-public-atom	NexusEngine Core
src/core-domains/object/subdomains/placement/kits/object-placement-kit/placement-math.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/subdomains/placement/kits/object-placement-kit/placement-operations.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/subdomains/placement/kits/object-placement-kit/placement-validation.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/subdomains/shape/index.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/subdomains/shape/kits/object-shape-kit/descriptors.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/subdomains/shape/kits/object-shape-kit/index.js	n:object	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
src/core-domains/object/subdomains/shape/kits/object-shape-kit/metrics.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/subdomains/shape/kits/object-shape-kit/profiles.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/subdomains/shape/kits/object-shape-kit/provider.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/subdomains/shape/kits/object-shape-kit/qualification.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/subdomains/shape/providers/meshoptimizer-shape-provider-kit/index.js	n:object	manifest-proven-public-atom	NexusEngine Core
src/core-domains/object/subdomains/shape/providers/meshoptimizer-shape-provider-kit/meshoptimizer-provider.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/subdomains/shape/providers/meshoptimizer-shape-provider-kit/reference-provider.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/subdomains/vegetation/adapters/vegetation-object-bridge-kit/index.js	n:object	manifest-proven-public-atom	NexusEngine Core
src/core-domains/object/subdomains/vegetation/index.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/subdomains/vegetation/kits/object-vegetation-kit/contracts.js	n:object	manifest-owned-internal	NexusEngine Core
src/core-domains/object/subdomains/vegetation/kits/object-vegetation-kit/index.js	n:object	manifest-proven-public-atom	NexusEngine Core
src/core-domains/object/subdomains/vegetation/subdomains/ecology-domain/index.js	n:object	manifest-proven-public-atom	NexusEngine Core
src/core-domains/object/subdomains/vegetation/subdomains/foilage-domain/index.js	n:object	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
src/core-domains/object/subdomains/vegetation/subdomains/tree-domain/index.js	n:object	manifest-proven-public-atom	NexusEngine Core
src/core-domains/policy/domain.manifest.js	n:policy	manifest-owned-internal	NexusEngine Core
src/core-domains/policy/index.js	n:policy	manifest-owned-internal	NexusEngine Core
src/core-domains/policy/kits/policy-kit/index.js	n:policy	manifest-proven-public-atom	NexusEngine Core
src/core-domains/presentation/contracts.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/domain.manifest.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/index.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/kits/presentation-kit/index.js	n:presentation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/presentation/subdomains/animation/kits/animation-kit/index.js	n:presentation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/presentation/subdomains/animation/kits/rig-transform-kit.js	n:presentation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/presentation/subdomains/audio/kits/audio-kit/index.js	n:presentation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/presentation/subdomains/camera/kits/camera-control-kit.js	n:presentation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/presentation/subdomains/camera/kits/camera-kit/index.js	n:presentation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/presentation/subdomains/camera/kits/camera-kit/smoothing.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/subdomains/camera/kits/framing-kit/index.js	n:presentation	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
src/core-domains/presentation/subdomains/camera/kits/framing-kit/orthographic-fit.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/subdomains/camera/kits/framing-kit/perspective-fit.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/subdomains/capture/index.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/subdomains/capture/kits/capture-kit/descriptors.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/subdomains/capture/kits/capture-kit/index.js	n:presentation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/presentation/subdomains/capture/kits/capture-kit/provider.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/subdomains/graphics/index.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/subdomains/graphics/kits/graphics-kit/adapters.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/subdomains/graphics/kits/graphics-kit/index.js	n:presentation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/presentation/subdomains/graphics/kits/graphics-kit/instance-batches.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/subdomains/graphics/kits/graphics-kit/lighting-descriptors.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/subdomains/graphics/kits/graphics-kit/material-descriptors.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/subdomains/graphics/kits/graphics-kit/procedural-material-descriptors.js	n:presentation	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
src/core-domains/presentation/subdomains/graphics/kits/graphics-kit/quality-profiles.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/subdomains/graphics/kits/graphics-kit/reflection-kit/contract.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/subdomains/graphics/kits/graphics-kit/reflection-kit/index.js	n:presentation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/presentation/subdomains/graphics/kits/graphics-kit/render-descriptors.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/subdomains/graphics/kits/graphics-kit/render-layer-graph-kit/contract.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/subdomains/graphics/kits/graphics-kit/render-layer-graph-kit/index.js	n:presentation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/presentation/subdomains/graphics/kits/graphics-kit/terrain-lod-descriptors.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/subdomains/graphics/kits/graphics-kit/vfx-descriptors.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/subdomains/output/kits/output-kit/index.js	n:presentation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/presentation/subdomains/output/kits/output-kit/math.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/subdomains/sky/kits/sky-kit/index.js	n:presentation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/presentation/subdomains/speech/index.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/presentation/subdomains/speech/kits/speech-kit/index.js	n:presentation	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
src/core-domains/presentation/subdomains/ui/kits/ui-kit/index.js	n:presentation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/presentation/subdomains/ui/kits/ui-scale-kit/index.js	n:presentation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/presentation/subdomains/ui/kits/ui-scale-kit/math.js	n:presentation	manifest-owned-internal	NexusEngine Core
src/core-domains/runtime/domain.manifest.js	n:runtime	manifest-owned-internal	NexusEngine Core
src/core-domains/runtime/index.js	n:runtime	manifest-owned-internal	NexusEngine Core
src/core-domains/runtime/kits/runtime-lifecycle-kit/index.js	n:runtime	manifest-proven-public-atom	NexusEngine Core
src/core-domains/runtime/subdomains/data/kits/data-kit/index.js	n:runtime	manifest-proven-public-atom	NexusEngine Core
src/core-domains/runtime/subdomains/data/kits/data-kit/ledger.js	n:runtime	manifest-owned-internal	NexusEngine Core
src/core-domains/runtime/subdomains/data/kits/data-kit/migration.js	n:runtime	manifest-owned-internal	NexusEngine Core
src/core-domains/runtime/subdomains/data/kits/data-kit/package-service.js	n:runtime	manifest-owned-internal	NexusEngine Core
src/core-domains/runtime/subdomains/data/kits/data-kit/schema.js	n:runtime	manifest-owned-internal	NexusEngine Core
src/core-domains/runtime/subdomains/data/kits/data-kit/selectors.js	n:runtime	manifest-owned-internal	NexusEngine Core
src/core-domains/runtime/subdomains/data/kits/data-kit/services.js	n:runtime	manifest-owned-internal	NexusEngine Core
src/core-domains/runtime/subdomains/data/kits/data-kit/snapshot.js	n:runtime	manifest-owned-internal	NexusEngine Core
src/core-domains/runtime/subdomains/persistence/kits/persistence-kit/index.js	n:runtime	manifest-proven-public-atom	NexusEngine Core
src/core-domains/runtime/subdomains/realtime/contracts/surfaces.js	n:runtime	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
src/core-domains/runtime/subdomains/realtime/contracts/tick-context-scheduler.js	n:runtime	manifest-owned-internal	NexusEngine Core
src/core-domains/runtime/subdomains/realtime/kits/realtime-kit/index.js	n:runtime	manifest-proven-public-atom	NexusEngine Core
src/core-domains/runtime/subdomains/sequence/kits/sequence-kit/index.js	n:runtime	manifest-proven-public-atom	NexusEngine Core
src/core-domains/runtime/subdomains/sequence/runtime/sequence-node-kit.js	n:runtime	manifest-owned-internal	NexusEngine Core
src/core-domains/runtime/subdomains/sequence/runtime/sequence-node-library.js	n:runtime	manifest-owned-internal	NexusEngine Core
src/core-domains/runtime/subdomains/sequence/runtime/sequence-node-runtime.js	n:runtime	manifest-owned-internal	NexusEngine Core
src/core-domains/runtime/subdomains/sequence/runtime/sequence-runtime.js	n:runtime	manifest-owned-internal	NexusEngine Core
src/core-domains/runtime/subdomains/startup/core-assets-startup-bridge.js	n:runtime	manifest-owned-internal	NexusEngine Core
src/core-domains/runtime/subdomains/startup/index.js	n:runtime	manifest-owned-internal	NexusEngine Core
src/core-domains/runtime/subdomains/startup/kits/startup-kit/index.js	n:runtime	manifest-proven-public-atom	NexusEngine Core
src/core-domains/runtime/subdomains/transaction/kits/transaction-ledger-kit/index.js	n:runtime	manifest-proven-public-atom	NexusEngine Core
src/core-domains/simulation/domain.manifest.js	n:simulation	manifest-owned-internal	NexusEngine Core
src/core-domains/simulation/index.js	n:simulation	manifest-owned-internal	NexusEngine Core
src/core-domains/simulation/kits/simulation-kit/checkpoints.js	n:simulation	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
src/core-domains/simulation/kits/simulation-kit/hazards.js	n:simulation	manifest-owned-internal	NexusEngine Core
src/core-domains/simulation/kits/simulation-kit/index.js	n:simulation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/simulation/kits/simulation-kit/objectives.js	n:simulation	manifest-owned-internal	NexusEngine Core
src/core-domains/simulation/kits/simulation-kit/pressure.js	n:simulation	manifest-owned-internal	NexusEngine Core
src/core-domains/simulation/kits/simulation-kit/resolution.js	n:simulation	manifest-owned-internal	NexusEngine Core
src/core-domains/simulation/kits/simulation-kit/resources.js	n:simulation	manifest-owned-internal	NexusEngine Core
src/core-domains/simulation/kits/simulation-kit/services.js	n:simulation	manifest-owned-internal	NexusEngine Core
src/core-domains/simulation/kits/simulation-kit/timers.js	n:simulation	manifest-owned-internal	NexusEngine Core
src/core-domains/simulation/subdomains/motion/index.js	n:simulation	manifest-owned-internal	NexusEngine Core
src/core-domains/simulation/subdomains/motion/kits/motion-kit/contracts.js	n:simulation	manifest-owned-internal	NexusEngine Core
src/core-domains/simulation/subdomains/motion/kits/motion-kit/index.js	n:simulation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/simulation/subdomains/motion/kits/two-bone-ik-kit/index.js	n:simulation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/simulation/subdomains/motion/subdomains/articulated-motion-domain/contracts.js	n:simulation	manifest-owned-internal	NexusEngine Core
src/core-domains/simulation/subdomains/motion/subdomains/articulated-motion-domain/index.js	n:simulation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/simulation/subdomains/physics/adapters/articulated-motion-drive-adapter/index.js	n:simulation	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
src/core-domains/simulation/subdomains/physics/index.js	n:simulation	manifest-owned-internal	NexusEngine Core
src/core-domains/simulation/subdomains/physics/kits/physics-kit/index.js	n:simulation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/simulation/subdomains/physics/kits/physics-kit/provider.js	n:simulation	manifest-owned-internal	NexusEngine Core
src/core-domains/simulation/subdomains/physics/subdomains/articulated-dynamics-domain/contracts.js	n:simulation	manifest-owned-internal	NexusEngine Core
src/core-domains/simulation/subdomains/physics/subdomains/articulated-dynamics-domain/index.js	n:simulation	manifest-proven-public-atom	NexusEngine Core
src/core-domains/spatial/domain.manifest.js	n:spatial	manifest-owned-internal	NexusEngine Core
src/core-domains/spatial/index.js	n:spatial	manifest-owned-internal	NexusEngine Core
src/core-domains/spatial/kits/angle-math-kit.js	n:spatial	manifest-proven-public-atom	NexusEngine Core
src/core-domains/spatial/kits/quaternion-math-kit.js	n:spatial	manifest-proven-public-atom	NexusEngine Core
src/core-domains/spatial/kits/spatial-kit/index.js	n:spatial	manifest-proven-public-atom	NexusEngine Core
src/core-domains/spatial/kits/transform-math-kit.js	n:spatial	manifest-proven-public-atom	NexusEngine Core
src/core-domains/spatial/kits/vector-math-kit.js	n:spatial	manifest-proven-public-atom	NexusEngine Core
src/core-domains/world/adapters/terrain-provider-adapter/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/domain.manifest.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/kits/world-builder-runtime-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
src/core-domains/world/kits/world-cell-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/kits/world-effect-provider-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/kits/world-partition-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/kits/world-surface-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/partitions/quad-tree-partition-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/partitions/uniform-grid-partition-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/portable.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/preparation/world-patch-preparation-controller.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/snapshot.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/state.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/scene/kits/scene-kit/constants.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/scene/kits/scene-kit/descriptors.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/scene/kits/scene-kit/host-contract.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/scene/kits/scene-kit/index.js	n:world	manifest-proven-public-atom	NexusEngine Core
src/core-domains/world/subdomains/scene/kits/scene-kit/ledgers.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/scene/kits/scene-kit/scene-registry.js	n:world	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
src/core-domains/world/subdomains/scene/kits/scene-kit/scene-transition-ledger.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/scene/kits/scene-kit/state.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/scene/kits/scene-kit/transitions.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/scene/kits/scene-kit/utils.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/weather/contracts.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/weather/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/weather/subdomains/layered-weather-domain/index.js	n:world	manifest-proven-public-atom	NexusEngine Core
src/core-domains/world/subdomains/weather/weather-domain.js	n:world	manifest-proven-public-atom	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/contracts.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/kits/feature-composition-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/kits/feature-definition-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/kits/feature-lifecycle-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/kits/feature-query-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
src/core-domains/world/subdomains/world-feature-domain/kits/feature-registry-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/kits/semantic-feature-kit/index.js	n:world	manifest-proven-public-atom	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/snapshot.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/state.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/subdomains/atmosphere-feature-domain/index.js	n:world	manifest-proven-public-atom	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/subdomains/ecology-feature-domain/index.js	n:world	manifest-proven-public-atom	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/subdomains/feature-family-domain-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/subdomains/hydrology-feature-domain/index.js	n:world	manifest-proven-public-atom	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/subdomains/landform-feature-domain/contracts.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/subdomains/landform-feature-domain/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/subdomains/landform-feature-domain/kits/canyon-feature-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
src/core-domains/world/subdomains/world-feature-domain/subdomains/landform-feature-domain/kits/cliff-feature-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/subdomains/landform-feature-domain/kits/landform-feature-kits.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/subdomains/landform-feature-domain/kits/mountain-feature-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/subdomains/landform-feature-domain/kits/plateau-feature-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/subdomains/landform-feature-domain/landform-feature-domain.js	n:world	manifest-proven-public-atom	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/subdomains/settlement-feature-domain/index.js	n:world	manifest-proven-public-atom	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/validation.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-feature-domain/world-feature-domain.js	n:world	manifest-proven-public-atom	NexusEngine Core
src/core-domains/world/subdomains/world-foundation-domain/contracts.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-foundation-domain/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-foundation-domain/kits/foundation-cell-resolution-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
src/core-domains/world/subdomains/world-foundation-domain/kits/foundation-composition-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-foundation-domain/kits/foundation-definition-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-foundation-domain/kits/foundation-sampling-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-foundation-domain/snapshot.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-foundation-domain/state.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-foundation-domain/validation.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/subdomains/world-foundation-domain/world-foundation-domain.js	n:world	manifest-proven-public-atom	NexusEngine Core
src/core-domains/world/surfaces/curved-horizon-surface-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/surfaces/flat-world-surface-kit/index.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/validation.js	n:world	manifest-owned-internal	NexusEngine Core
src/core-domains/world/world-domain.js	n:world	manifest-proven-public-atom	NexusEngine Core
src/domain-api.js	engine-root	root-contract	NexusEngine minimal root
src/domain-path.js	engine-root	root-contract	NexusEngine minimal root
src/domain-service-kit.js	engine-root	root-contract	NexusEngine minimal root
src/ecs.js	engine-root	root-contract	NexusEngine minimal root
src/engine.js	engine-root	root-contract	NexusEngine minimal root
src/foundation/completion-ledger.js	engine-foundation	root-contract	NexusEngine foundation
src/foundation/deterministic-replay.js	engine-foundation	root-contract	NexusEngine foundation

Path	Owner	Review	Destination
src/foundation/idempotency- ledger.js	engine-foundation	root-contract	NexusEngine foundation
src/foundation/index.js	engine-foundation	root-contract	NexusEngine foundation
src/foundation/progress- timer.js	engine-foundation	root-contract	NexusEngine foundation
src/foundation/promotion- guard.js	engine-foundation	root-contract	NexusEngine foundation
src/foundation/seeded- random.js	engine-foundation	root-contract	NexusEngine foundation
src/foundation/serializable- state.js	engine-foundation	root-contract	NexusEngine foundation
src/foundation/sha256.js	engine-foundation	root-contract	NexusEngine foundation
src/foundation/snapshot.js	engine-foundation	root-contract	NexusEngine foundation
src/index.js	engine-root	root-contract	NexusEngine minimal root
src/release.js	engine-root	root-contract	NexusEngine minimal root
src/runtime-kit.js	engine-root	root-contract	NexusEngine minimal root

GENERATED REFERENCE

NexusEngine 0.0.4 Root Module Dispositions

Source commit: `a9adca5b3620f996f00860358c4864dd4bdfa6d9`

This is a hard cutover. Removed modules are not forwarded by NexusEngine. Rebuild them from the named Core atoms in the owning package or game.

Removed Exports From Retained Modules

Source	Removed exports	Disposition	Replacement
<code>src/core-domains/compute/subdomains/model/kits/model-kit/index.js</code>	<code>createMockModelAdapter</code>	recipe-data	Inject <code>createModelAdapterBoundary({ infer })</code> into <code>createModelKit({ adapters })</code> .
<code>src/core-domains/presentation/subdomains/sky/kits/sky-kit/index.js</code>	<code>CORE_SKYBOX_PRESETS</code>	recipe-data	Pass caller-owned preset data to <code>createSkyDescriptorKit({ presets })</code> or <code>registerPreset()</code> .
<code>src/core-domains/presentation/subdomains/speech/kits/speech-kit/index.js</code>	<code>createTinyTTSAAssetManifest</code> , <code>createTinyTTSPProvider</code> , <code>registerTinyTTSAAssets</code>	external-kit	Install a model, network, and asset implementation through <code>createSpeechKit().registerProvider()</code> .
<code>src/core-domains/presentation/subdomains/speech/kits/speech-kit/pocket-tts-provider.js</code>	<code>createPocketTTSAAssetManifest</code> , <code>createPocketTTSHttpAssetProvider</code> , <code>createPocketTTSPProvider</code> , <code>registerPocketTTSAAssets</code>	external-kit	Install PocketTTS through <code>createSpeechKit().registerProvider()</code> .

Retired Source Modules

Source	Disposition	Owner	Core requirements
<code>bin/nexus-editor.mjs</code>	external-kit	NexusEngine-Editor	<code>n:composition</code> , <code>n:host</code> , <code>n:mcp</code>
<code>src/action-movement-kit.js</code>	external-kit	NexusEngine-Kits	<code>n:actor</code> , <code>n:interaction:input</code> , <code>n:simulation:motion</code>
<code>src/assistance-target-kit.js</code>	external-kit	NexusEngine-Kits	<code>n:actor</code> , <code>n:interaction</code> , <code>n:object</code> , <code>n:simulation</code>
<code>src/cargo-manifest-kit.js</code>	external-kit	NexusEngine-Kits	<code>n:actor</code> , <code>n:object</code> , <code>n:runtime:data</code> , <code>n:runtime:transaction</code>
<code>src/character-camera-kit.js</code>	external-kit	NexusEngine-Kits	<code>n:actor:character</code> , <code>n:presentation:camera</code> , <code>n:spatial</code>
<code>src/common-game-definitions.js</code>	recipe-data	NexusEngine-Kits recipes	<code>n:actor</code> , <code>n:interaction</code> , <code>n:simulation</code>
<code>src/core-domains/compute/subdomains/model/kits/model-kit/mock-model-adapter.js</code>	recipe-data	NexusEngine tests	<code>n:compute:model</code>

Source	Disposition	Owner	Core requirements
src/core-domains/presentation/subdomains/speech/kits/speech-kit/pocket-tts-provider.js	external-kit	NexusEngine-Kits adapters	n:asset , n:compute:model , n:presentation:speech
src/core-domains/world/subdomains/scene/kits/scene-kit/hosts/headless-scene-host.js	external-kit	NexusEngine-Editor adapters	n:host , n:world:scene
src/core-domains/world/subdomains/scene/kits/scene-kit/hosts/native-scene-host.js	external-kit	NexusEngine-Editor adapters	n:host , n:world:scene
src/core-domains/world/subdomains/scene/kits/scene-kit/hosts/rust-native-scene-host.js	external-kit	NexusEngine-Editor adapters	n:host , n:world:scene
src/core-domains/world/subdomains/scene/kits/scene-kit/hosts/web-html-scene-host.js	external-kit	NexusEngine-Editor adapters	n:host , n:world:scene
src/core-domains/world/subdomains/scene/kits/scene-kit/hosts/web-scene-host.js	external-kit	NexusEngine-Editor adapters	n:host , n:world:scene
src/economy-kit.js	external-kit	NexusEngine-Kits	n:runtime:data , n:runtime:transaction , n:simulation
src/environmental-affordance-kit.js	external-kit	NexusEngine-Kits	n:interaction , n:object , n:spatial
src/facility-operations-kit.js	external-kit	NexusEngine-Kits	n:runtime:sequence , n:runtime:transaction , n:simulation
src/game-kit-composer.js	duplicate	n:composition	n:composition , n:policy
src/hazard-field-kit.js	external-kit	NexusEngine-Kits	n:object , n:simulation , n:spatial
src/host.js	external-kit	NexusEngine-Editor	n:composition , n:host , n:policy
src/hosts/browser/browser-indexeddb-asset-cache-adapter.js	external-kit	NexusEngine-Editor adapters	n:asset , n:host , n:presentation
src/hosts/browser/browser-presentation-surface-adapter.js	external-kit	NexusEngine-Editor adapters	n:asset , n:host , n:presentation
src/hosts/browser/browser-startup-presentation-adapter.js	external-kit	NexusEngine-Editor adapters	n:asset , n:host , n:presentation
src/landmark-guidance-kit.js	external-kit	NexusEngine-Kits	n:object , n:spatial , n:world
src/lifecycle-progression-kit.js	external-kit	NexusEngine-Kits	n:runtime:sequence , n:runtime:transaction , n:simulation

Source	Disposition	Owner	Core requirements
src/modules/nexus-diffusion/index.js	external-kit	NexusEngine-Kits	n:compute, n:compute:model, n:runtime:data
src/modules/nexus-diffusion/nexus-diffusion-domain.js	external-kit	NexusEngine-Kits	n:compute, n:compute:model, n:runtime:data
src/modules/nexus-diffusion/subdomains/backend/index.js	external-kit	NexusEngine-Kits	n:compute, n:compute:model, n:runtime:data
src/modules/nexus-diffusion/subdomains/checkpoint/index.js	external-kit	NexusEngine-Kits	n:compute, n:compute:model, n:runtime:data
src/modules/nexus-diffusion/subdomains/dataset/index.js	external-kit	NexusEngine-Kits	n:compute, n:compute:model, n:runtime:data
src/modules/nexus-diffusion/subdomains/model/index.js	external-kit	NexusEngine-Kits	n:compute, n:compute:model, n:runtime:data
src/modules/nexus-diffusion/subdomains/noise/index.js	external-kit	NexusEngine-Kits	n:compute, n:compute:model, n:runtime:data
src/modules/nexus-diffusion/subdomains/preview/index.js	external-kit	NexusEngine-Kits	n:compute, n:compute:model, n:runtime:data
src/modules/nexus-diffusion/subdomains/sampling/index.js	external-kit	NexusEngine-Kits	n:compute, n:compute:model, n:runtime:data
src/modules/nexus-diffusion/subdomains/tensor/index.js	external-kit	NexusEngine-Kits	n:compute, n:compute:model, n:runtime:data
src/modules/nexus-diffusion/subdomains/training/index.js	external-kit	NexusEngine-Kits	n:compute, n:compute:model, n:runtime:data
src/modules/nexus-diffusion/utils/seeded-random.js	external-kit	NexusEngine-Kits	n:compute, n:compute:model, n:runtime:data
src/navmesh-kit.js	external-kit	NexusEngine-Kits	n:compute, n:spatial, n:world
src/occupant-flow-kit.js	external-kit	NexusEngine-Kits	n:actor, n:runtime:sequence, n:simulation
src/pathfinding-kit.js	external-kit	NexusEngine-Kits	n:compute, n:spatial, n:world
src/procedural-kit.js	external-kit	NexusEngine-Kits	n:compute, n:runtime:data, n:world
src/procedural-navigation-extensions.js	duplicate	NexusEngine-Kits	n:composition
src/pursuit-pressure-kit.js	external-kit	NexusEngine-Kits	n:actor, n:simulation, n:spatial
src/renderers.js	external-kit	NexusEngine-Kits adapters	n:host, n:presentation:output

Source	Disposition	Owner	Core requirements
src/renderers/three/three-object-capture-provider.js	external-kit	NexusEngine-Kits adapters	n:host , n:object , n:presentation
src/renderers/three/three-presentation-output-adapter.js	external-kit	NexusEngine-Kits adapters	n:host , n:object , n:presentation
src/renderers/three/three-subject-bounds-adapter.js	external-kit	NexusEngine-Kits adapters	n:host , n:object , n:presentation
src/request-fulfillment-kit.js	external-kit	NexusEngine-Kits	n:interaction , n:runtime:sequence , n:simulation
src/request-queue-kit.js	external-kit	NexusEngine-Kits	n:actor , n:runtime:sequence , n:runtime:transaction
src/route-field-kit.js	external-kit	NexusEngine-Kits	n:object , n:spatial , n:world
src/scenario-driver-kit.js	game-owned	Experiments or game repository	n:composition , n:runtime:sequence , n:simulation
src/scenario-duration-kit.js	game-owned	Experiments or game repository	n:runtime:sequence , n:simulation
src/schedule-kit.js	external-kit	NexusEngine-Kits	n:runtime:sequence , n:simulation
src/shaders.js	external-kit	NexusEngine-Kits adapters	n:host , n:presentation:graphics
src/spatial-scale-kit.js	external-kit	NexusEngine-Kits	n:object , n:spatial
src/terrain-kit.js	external-kit	NexusEngine-Kits	n:compute , n:world , n:world:scene
src/transfer-zone-kit.js	external-kit	NexusEngine-Kits	n:interaction , n:object , n:spatial
src/transport-route-kit.js	external-kit	NexusEngine-Kits	n:actor , n:runtime:sequence , n:world
src/vehicle-dynamics-kit.js	external-kit	NexusEngine-Kits	n:interaction:input , n:simulation:motion , n:simulation:physics
src/water-surface-kit.js	external-kit	NexusEngine-Kits	n:presentation:graphics , n:world
src/world-physics-kit.js	external-kit	NexusEngine-Kits	n:simulation:motion , n:simulation:physics , n:world
tools/headless-editor/	external-kit	NexusEngine-Editor	n:composition , n:host , n:mcp , n:policy

GENERATED REFERENCE

Frozen ProtoKit Extraction

Canonical source: <https://github.com/LuminaryLabs-Agents/NexusEngine-ProtoKits.git> at `0d102649267737230d618b30fe6f9465b198d234`

Coverage: **3698/3698 source items (100%)** across **530 ProtoKit folders**. Source code was parsed as text and never executed.

Dispositions

Disposition	Source items
core-reuse	119
core-composition	115
core-new-atom	0
external-kit	3096
recipe-data	113
game-owned	236
duplicate	0
rejected-unproven	19

Core Gap Set

No historical source item passed every Core-new-atom gate. The frozen Core gap set is empty; unproven generic-looking behavior remains external or rejected rather than being promoted by name.

Deterministic Batches

Batch	First folder	Last folder	Folders
batch-01	2d-platformer-domain	blackwake-kit-registry	50
batch-02	blackwake-preset-kit	domain-taxonomy-domain-kit	50
batch-03	downhill-race-kit	generic-cargo-kit	50
batch-04	generic-cargo-transfer-kit	generic-replay-test-kit	50
batch-05	generic-resource-loop-kit	ground-contact-kit	50
batch-06	guard-domain-kit	navigation-knowledge-domain-kit	50
batch-07	negotiation-commitment-domain-kit	openxr-hand-adapter-dsk	50
batch-08	orchard-biome-kit	render-layer-kit	50
batch-09	render-quality-budget-kit	stream-backpressure-kit	50
batch-10	stream-channel-kit	water-stream-kit	50
batch-11	water-surface-kit	zone-field-kit	30

Artifacts

- `source.json`: immutable source identity and complete snapshot hash.

- `inventory.json` : files, exports, state/lifecycle signals, dependencies, consumers, providers, and adapters by folder.
- `dispositions.json` : canonical export-level ownership and reconstruction ledger.
- `core-gap-set.json` : frozen result of the new-Core-atom gate.